

**RAMCO INSTITUTE OF TECHNOLOGY**

**DEPARTMENT OF ELECTRONICS AND  
COMMUNICATION ENGINEERING**

**COURSE OUTCOMES**

**REGULATION 2013**

# **CRITERIA 3**

## Course Outcomes

### 3.1.1. Course Outcomes (COs)(05)

(SAR should include course outcomes of one course from each semester of study, however, should be prepared for all courses and made available as evidence, if asked)

The following are course outcomes based on Regulation 2013.

| <b>COURSE OUTCOMES – BATCH 2015-2019</b>                                                                                                               |                                                                                                                                                              |
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| <b>I Year</b>                                                                                                                                          |                                                                                                                                                              |
| <b>Course Code &amp; Title : HS6151 Technical English - I</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C101</b>   |                                                                                                                                                              |
| <b>Couse Index</b>                                                                                                                                     | <b>Course Outcomes</b>                                                                                                                                       |
| C101.1                                                                                                                                                 | Explain clearly, confidently, comprehensibly, and communicate with one or many listeners using appropriate communicative strategies.                         |
| C101.2                                                                                                                                                 | Construct cohesively and coherently and flawlessly avoiding grammatical errors, using a wide vocabulary range, organizing their ideas logically on a topic.  |
| C101.3                                                                                                                                                 | Organize different genres of texts adopting various reading strategies.                                                                                      |
| C101.4                                                                                                                                                 | Distinguish and comprehend different spoken discourses/excerpts in different accents.                                                                        |
| C101.5                                                                                                                                                 | Improve listening to different accents, speeches and presentations.                                                                                          |
| <b>Course Code &amp; Title : MA6151 Mathematics - I</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C102</b>         |                                                                                                                                                              |
| <b>Couse Index</b>                                                                                                                                     | <b>Course Outcomes</b>                                                                                                                                       |
| C102.1                                                                                                                                                 | Find canonical form from quadratic form using the concepts of Eigen values and Eigen vectors.                                                                |
| C102.2                                                                                                                                                 | Test the convergence of sequences and series.                                                                                                                |
| C102.3                                                                                                                                                 | Apply the techniques of differential calculus to find the evaluate and envelope of curves.                                                                   |
| C102.4                                                                                                                                                 | Determine the maxima and minima of functions of two variables using partial derivatives.                                                                     |
| C102.5                                                                                                                                                 | Find the area enclosed by plane curves and volume of solids using multiple integrals.                                                                        |
| <b>Course Code &amp; Title : PH6151 Engineering Physics - I</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C103</b> |                                                                                                                                                              |
| <b>Couse Index</b>                                                                                                                                     | <b>Course Outcomes</b>                                                                                                                                       |
| C103.1                                                                                                                                                 | Extend the knowledge about the crystal structures and crystal growth techniques.                                                                             |
| C103.2                                                                                                                                                 | Interpret the fundamental knowledge about the elastic nature of materials and able to differentiate a good thermal conductor from the bad thermal conductor. |

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| C103.3                                                                                                                                                   | Apply the knowledge of quantum mechanics and classical mechanics in addressing problems related to science and technology.                             |
| C103.4                                                                                                                                                   | Utilize the knowledge about designing an auditorium with good acoustical properties and make use of ultrasonic and its applications in various fields. |
| C103.5                                                                                                                                                   | Identify and appreciate the advantages of optical communication using LASER.                                                                           |
| <b>Course Code &amp; Title : CY6151 Engineering Chemistry - I</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C104</b> |                                                                                                                                                        |
| <b>Couse Index</b>                                                                                                                                       | <b>Course Outcomes</b>                                                                                                                                 |
| C104.1                                                                                                                                                   | Describe the methods of polymerization, types and properties of polymers.                                                                              |
| C104.2                                                                                                                                                   | Illustrate the concepts of basic thermodynamics and problem solving skills in various disciplines of Engineering.                                      |
| C104.3                                                                                                                                                   | Discuss the laws of photochemistry in recognizing the interaction of light with matter and its applications in luminescence and spectroscopy.          |
| C104.4                                                                                                                                                   | Review the use of the phase rule in identifying its applications in Metallurgical Engineering.                                                         |
| C104.5                                                                                                                                                   | Summarize the basic knowledge in Nano chemistry and distinguish the existing technology with Nano technology.                                          |
| <b>Course Code &amp; Title : GE6151 Computer Programming</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C105</b>      |                                                                                                                                                        |
| <b>Couse Index</b>                                                                                                                                       | <b>Course Outcomes</b>                                                                                                                                 |
| C105.1                                                                                                                                                   | Describe the functions of a digital computer with its organization and translate program logic into pseudo code and flow chart                         |
| C105.2                                                                                                                                                   | Apply the different conditional constructs to solve simple scientific and statistical problems                                                         |
| C105.3                                                                                                                                                   | Demonstrate the usage of functions and pointers with the help of C programs                                                                            |
| C105.4                                                                                                                                                   | Use the arrays and strings for writing programs involving character manipulations                                                                      |
| C105.5                                                                                                                                                   | Apply the concept of structures and unions in writing C programs                                                                                       |
| <b>Course Code &amp; Title : GE6152 Engineering Graphics</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C106</b>      |                                                                                                                                                        |
| <b>Couse Index</b>                                                                                                                                       | <b>Course Outcomes</b>                                                                                                                                 |
| C106.1                                                                                                                                                   | Familiarize with the fundamentals and standards of Engineering graphics                                                                                |
| C106.2                                                                                                                                                   | Perform freehand sketching of basic geometrical constructions and multiple views of objects.                                                           |

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| C106.3                                                                                                                                                          | Project orthographic projections of lines and plane surfaces.                                             |
| C106.4                                                                                                                                                          | Draw projections and solids and development of surfaces.                                                  |
| C106.5                                                                                                                                                          | The student will be able to Visualize and to project isometric and perspective sections of simple solids. |
| <b>Course Code &amp; Title : GE6161 Computer Practices Laboratory</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C107</b>    |                                                                                                           |
| <b>Couse Index</b>                                                                                                                                              | <b>Course Outcomes</b>                                                                                    |
| C107.1                                                                                                                                                          | Use of office automation tools.                                                                           |
| C107.2                                                                                                                                                          | Apply good programming design methods for program development.                                            |
| C107.3                                                                                                                                                          | Design and implement the C programs for simple applications.                                              |
| C107.4                                                                                                                                                          | Develop and implement the recursive programs.                                                             |
| C107.5                                                                                                                                                          | Implement the C programs with the help of structures and unions.                                          |
| <b>Course Code &amp; Title : GE6162 Engineering Practices Laboratory</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C108</b> |                                                                                                           |
| <b>Couse Index</b>                                                                                                                                              | <b>Course Outcomes</b>                                                                                    |
| C108.1                                                                                                                                                          | Construct carpentry components and pipe connections including plumbing works.                             |
| C108.2                                                                                                                                                          | Use welding equipment's to join the structures                                                            |
| C108.3                                                                                                                                                          | Illustrate the basic machining operations                                                                 |
| C108.4                                                                                                                                                          | Construct the models using sheet metal works                                                              |
| C108.5                                                                                                                                                          | Describe centrifugal pump, Air conditioner, operations of smithy, foundry and fittings.                   |
| C108.6                                                                                                                                                          | Construct the basic Electrical and Electronics circuits.                                                  |
| C108.7                                                                                                                                                          | Examine the different types of electronic circuits and components.                                        |
| C108.8                                                                                                                                                          | Explain the electrical safety rules, grounding, general house wiring.                                     |
| C108.9                                                                                                                                                          | Perform soldering in various electronic circuits.                                                         |
| C108.10                                                                                                                                                         | Illustrate the basic operation of domestic electrical appliances.                                         |

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| <b>Course Code &amp; Title : GE6163 Physics and Chemistry Laboratory - I</b><br><b>Semester &amp; Year of study : I &amp; 2015-16</b><br><b>Course Index : C109</b> |                                                                                                                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                  | <b>Course Outcomes</b>                                                                                                                                                                              |
| C109.1                                                                                                                                                              | Experienced with hands-on knowledge in the quantitative chemical analysis of chloride and dissolved oxygen.                                                                                         |
| C109.2                                                                                                                                                              | Equipped in the determination of acids by using the instruments of conductivity meter and pH meter.                                                                                                 |
| <b>Course Code &amp; Title : HS6251 Technical English – II</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C110</b>              |                                                                                                                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                  | <b>Course Outcomes</b>                                                                                                                                                                              |
| C110.1                                                                                                                                                              | Speak convincingly, express their opinions clearly, initiate a discussion, negotiate, argue using appropriate communicative strategies.                                                             |
| C110.2                                                                                                                                                              | Write effectively and persuasively and produce different types of writing such as narration, description, exposition and argument as well as creative, critical, analytical and evaluative writing. |
| C110.3                                                                                                                                                              | Read different genres of texts, infer implied meanings and critically analyse and evaluate them for ideas as well as for method of presentation.                                                    |
| C110.4                                                                                                                                                              | Listen/view and comprehend different spoken excerpts critically and infer unspoken and implied meanings.                                                                                            |
| C110.5                                                                                                                                                              | Students will become accomplished, active readers and will be able to write effectively for a variety of professional and social settings                                                           |
| <b>Course Code &amp; Title : MA6251 Mathematics– II</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C111</b>                     |                                                                                                                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                  | <b>Course Outcomes</b>                                                                                                                                                                              |
| C111.1                                                                                                                                                              | Apply knowledge of vector calculus in engineering disciplines                                                                                                                                       |
| C111.2                                                                                                                                                              | Solve ordinary differential equations that model the engineering problems                                                                                                                           |
| C111.3                                                                                                                                                              | Find the Laplace transform of functions and solve the ordinary differential equations using Laplace transform                                                                                       |
| C111.4                                                                                                                                                              | Construct analytic functions and apply the knowledge of conformal mappings in engineering disciplines                                                                                               |
| C111.5                                                                                                                                                              | Evaluate contour integration and apply it in engineering problems                                                                                                                                   |
| <b>Course Code &amp; Title : PH6251 Engineering Physics – II</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C112</b>            |                                                                                                                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                  | <b>Course Outcomes</b>                                                                                                                                                                              |
| C112.1                                                                                                                                                              | Describe the conducting materials and their properties.                                                                                                                                             |

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| C112.2                                                                                                                                                    | Analyze the semiconductors and able to differentiate various types of semiconductors.                                          |
| C112.3                                                                                                                                                    | Apply the knowledge of magnetic and superconducting materials for modern day to day applications.                              |
| C112.4                                                                                                                                                    | Explain the properties and applications of dielectrics.                                                                        |
| C112 .5                                                                                                                                                   | Apply the knowledge about the modern engineering materials for various applications.                                           |
| <b>Course Code &amp; Title : CY6251 Engineering Chemistry –II</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C113</b> |                                                                                                                                |
| <b>Couse Index</b>                                                                                                                                        | <b>Course Outcomes</b>                                                                                                         |
| C113.1                                                                                                                                                    | Describe water technology in the purification of water in domestic and industrial applications.                                |
| C113.2                                                                                                                                                    | Explain the principles of electrochemistry, the factors affecting corrosion and the prevention of corrosion.                   |
| C113.3                                                                                                                                                    | Classify the different alternative sources of energy and the generation processes.                                             |
| C113.4                                                                                                                                                    | Enumerate the different types of engineering materials and their applications.                                                 |
| C113.5                                                                                                                                                    | Discuss the industrial techniques of petroleum processing and the determination of calorific values and combustion parameters. |
| <b>Course Code &amp; Title : EC6201 Electronic Devices</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C114</b>        |                                                                                                                                |
| <b>Couse Index</b>                                                                                                                                        | <b>Course Outcomes</b>                                                                                                         |
| C114.1                                                                                                                                                    | Design applications using the basic electronic devices                                                                         |
| C114.2                                                                                                                                                    | Interpret the construction and operation of Bipolar Junction Transistor                                                        |
| C114.3                                                                                                                                                    | Analyze the operation of various Field Effect Transistors                                                                      |
| C114.4                                                                                                                                                    | Develop applications using the special semiconductor devices                                                                   |
| C114.5                                                                                                                                                    | Apply the basic concepts of Power and Display devices                                                                          |
| <b>Course Code &amp; Title : EE6201 Circuit Theory</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C115</b>            |                                                                                                                                |
| <b>Couse Index</b>                                                                                                                                        | <b>Course Outcomes</b>                                                                                                         |

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| C115.1                                                                                                                                                                         | Analyse circuits with ideal, independent, and controlled voltage and current sources.                                                               |
| C115.2                                                                                                                                                                         | Apply Ohm's Law, Kirchhoff's laws, Network theorems to the analysis of AC and DC circuits.                                                          |
| C115.3                                                                                                                                                                         | Evaluate the response of the series and parallel resonance circuits, coupled circuits and tuned circuits.                                           |
| C115.4                                                                                                                                                                         | Solve first and second order AC and DC circuits for steady-state and transient response in the time domain and two port networks.                   |
| C115.5                                                                                                                                                                         | Estimate the power consumed by the three-phase balanced and unbalanced circuits.                                                                    |
| <b>Course Code &amp; Title : GE6262 Physics and Chemistry Laboratory – II</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C116</b>          |                                                                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                             | <b>Course Outcomes</b>                                                                                                                              |
| C116.1                                                                                                                                                                         | Test materials by using their knowledge of applied physics principles in optics and properties of matter.                                           |
| C116.2                                                                                                                                                                         | Determine the hardness, alkalinity and metal ion content in the water samples by volumetric titration.                                              |
| C116.3                                                                                                                                                                         | Estimate the water quality parameters by potentiometer, conductometer and flame photometer.                                                         |
| <b>Course Code &amp; Title : EC6211 Circuits and Devices Laboratory</b><br><b>Semester &amp; Year of study : II &amp; 2015-16</b><br><b>Course Index : C117</b>                |                                                                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                             | <b>Course Outcomes</b>                                                                                                                              |
| C117.1                                                                                                                                                                         | Analyze the characteristics of basic electronic devices.                                                                                            |
| C117.2                                                                                                                                                                         | Design the RL & RC circuits.                                                                                                                        |
| C117.3                                                                                                                                                                         | Apply KVL, KCL, Thevenin, Norton and Super Position Theorems in circuit analysis.                                                                   |
| <b>II Year</b>                                                                                                                                                                 |                                                                                                                                                     |
| <b>Course Code &amp; Title : MA6351 Transforms and Partial Differential Equations</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C201</b> |                                                                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                             | <b>Course Outcomes</b>                                                                                                                              |
| C201.1                                                                                                                                                                         | Solve the Partial Differential Equations.                                                                                                           |
| C201.2                                                                                                                                                                         | Determine the Fourier series expansion of functions and hence evaluate the value of infinite series.                                                |
| C201.3                                                                                                                                                                         | Apply the method of separation of variables to solve one dimensional wave equation, one dimensional heat equation and two dimensional heat equation |



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| C201.4                                                                                                                                                                           | Find the Fourier transform of functions and also evaluate definite integrals using Fourier transform.             |
| C201.5                                                                                                                                                                           | Calculate the Z-transform of discrete time systems and solve the difference equations using Z-transform.          |
| <b>Course Code &amp; Title : EE6352 Electrical Engineering and Instrumentation</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C202</b>      |                                                                                                                   |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                                            |
| C202.1                                                                                                                                                                           | Analyze the characteristics of DC machines                                                                        |
| C202.2                                                                                                                                                                           | Determine the efficiency and regulation of transformers                                                           |
| C202.3                                                                                                                                                                           | Evaluate the performance of Induction and Synchronous machines                                                    |
| C202.4                                                                                                                                                                           | Select transducers for the measurement of temperature, displacement and strain                                    |
| C202.5                                                                                                                                                                           | Determine the circuit parameters using AC and DC bridges                                                          |
| <b>Course Code &amp; Title : EC6301 Object Oriented Programming and Data Structures</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C203</b> |                                                                                                                   |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                                            |
| C203.1                                                                                                                                                                           | Summarize the object oriented features like data abstraction and overloading to solve various computing problems. |
| C203.2                                                                                                                                                                           | Illustrate the concepts of inheritance and polymorphism in various applications.                                  |
| C203.3                                                                                                                                                                           | Make use of linear data structure concepts for problem solving.                                                   |
| C203.4                                                                                                                                                                           | Experiment with various non-linear data structures such as trees and graphs.                                      |
| C203.5                                                                                                                                                                           | Analyze standard algorithms for searching and sorting techniques.                                                 |
| <b>Course Code &amp; Title : EC6302 Digital Electronics</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C204</b>                             |                                                                                                                   |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                                            |
| C204.1                                                                                                                                                                           | Analyze different methods used for simplification of Boolean functions                                            |
| C204.2                                                                                                                                                                           | Design and implement Combinational circuits                                                                       |
| C204.3                                                                                                                                                                           | Design and implement synchronous and asynchronous sequential circuits                                             |

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| C204.4                                                                                                                                                                  | Apply the basic concepts of memory devices and programmable logic devices            |
| C204.5                                                                                                                                                                  | Write simple HDL code for combinational and sequential circuits                      |
| <b>Course Code &amp; Title : EC6303 Signals and Systems</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C205</b>                    |                                                                                      |
| <b>Couse Index</b>                                                                                                                                                      | <b>Course Outcomes</b>                                                               |
| C205.1                                                                                                                                                                  | Analyze the various properties of signals and systems.                               |
| C205.2                                                                                                                                                                  | Apply Laplace transform, Fourier transform, Z transform and DTFT in signal analysis. |
| C205.3                                                                                                                                                                  | Analyse continuous time LTI systems using Fourier and Laplace transform.             |
| C205.4                                                                                                                                                                  | Analyze the sampling theorem and discrete time signals using Z transform and DTFT.   |
| C205.5                                                                                                                                                                  | Interpret the LTI discrete time systems using Z transform and DTFT                   |
| <b>Course Code &amp; Title : EC6304 Electronic Circuits- I</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C206</b>                 |                                                                                      |
| <b>Couse Index</b>                                                                                                                                                      | <b>Course Outcomes</b>                                                               |
| C206.1                                                                                                                                                                  | Design the circuits with BJT and MOSFET biasing                                      |
| C206.2                                                                                                                                                                  | Analyze the small signal equivalent and design simple BJT amplifier circuits         |
| C206.3                                                                                                                                                                  | Design simple JFET and MOSFET amplifier circuits                                     |
| C206.4                                                                                                                                                                  | Find the frequency response of all amplifiers                                        |
| C206.5                                                                                                                                                                  | Design an amplifiers with active loads                                               |
| <b>Course Code &amp; Title : EC6311 Analog and Digital Circuits Laboratory</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C207</b> |                                                                                      |
| <b>Couse Index</b>                                                                                                                                                      | <b>Course Outcomes</b>                                                               |
| C207.1                                                                                                                                                                  | Design the electronic circuits.                                                      |
| C207.2                                                                                                                                                                  | Recognize the characteristics of CE,CB& CC amplifiers                                |
| C207.3                                                                                                                                                                  | Calculate the frequency response of CS amplifier.                                    |

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| C207.4                                                                                                                                                               | Analyze the limitations in bandwidth of single stage and multistage amplifiers.                                                           |
| C207.5                                                                                                                                                               | Simulate the amplifiers using spice tool.                                                                                                 |
| C207.6                                                                                                                                                               | Design and implement the digital circuits using logic gates.                                                                              |
| <b>Course Code &amp; Title : EC6312 OOPS and Data Structures Laboratory</b><br><b>Semester &amp; Year of study : III &amp; 2016-17</b><br><b>Course Index : C208</b> |                                                                                                                                           |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                                    |
| C208.1                                                                                                                                                               | Design and implement various concepts in C++ programming language.                                                                        |
| C208.2                                                                                                                                                               | Apply good programming design methods for program development.                                                                            |
| C208.3                                                                                                                                                               | Apply the different data structures like stacks, queues, linked lists, trees and graphs for implementing solutions to practical problems. |
| <b>Course Code &amp; Title : MA6451 Probability and Random Processes</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C209</b>     |                                                                                                                                           |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                                    |
| C209.1                                                                                                                                                               | Apply the knowledge of the probability concepts and distributions in communication engineering.                                           |
| C209.2                                                                                                                                                               | Find the covariance, correlation and regression of two dimensional random variables.                                                      |
| C209.3                                                                                                                                                               | Analyze the random processes to apply in random signals in communication engineering.                                                     |
| C209.4                                                                                                                                                               | Apply the concept of correlation functions and spectral densities of random processes in communication engineering problems               |
| C209.5                                                                                                                                                               | Analyze the response of random inputs to linear time invariant systems to apply in linear systems in communication engineering.           |
| <b>Course Code &amp; Title : EC6401 Electronic Circuits II</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C210</b>               |                                                                                                                                           |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                                    |
| C210.1                                                                                                                                                               | Design and analyze feedback amplifiers.                                                                                                   |
| C210.2                                                                                                                                                               | Design different types of oscillator at different frequency.                                                                              |
| C210.3                                                                                                                                                               | Analyse the performance of Tuned amplifiers.                                                                                              |
| C210.4                                                                                                                                                               | Design wave shaping circuits and multivibrators.                                                                                          |

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| C210.5                                                                                                                                                     | Apply the various design techniques to analyze blocking oscillators and time base Generators.                                        |
| <b>Course Code &amp; Title : EC6402 Communication Theory</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C211</b>       |                                                                                                                                      |
| <b>Couse Index</b>                                                                                                                                         | <b>Course Outcomes</b>                                                                                                               |
| C211.1                                                                                                                                                     | Analyze the various modulation techniques used for communication.                                                                    |
| C211.2                                                                                                                                                     | Elaborate the angle modulation and demodulation techniques.                                                                          |
| C211.3                                                                                                                                                     | Apply the concepts of Random process.                                                                                                |
| C211.4                                                                                                                                                     | Analyze the noise performance of AM and FM systems.                                                                                  |
| C211.5                                                                                                                                                     | Apply the basic concepts of source and channel coding in design of communication systems.                                            |
| <b>Course Code &amp; Title : EC6403 Electromagnetic Fields</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C212</b>     |                                                                                                                                      |
| <b>Couse Index</b>                                                                                                                                         | <b>Course Outcomes</b>                                                                                                               |
| C212.1                                                                                                                                                     | Apply the basic mathematical concepts of vector analysis and its associated laws to the solution of electrostatic field.             |
| C212.2                                                                                                                                                     | Describe the properties of conductors and dielectrics.                                                                               |
| C212.3                                                                                                                                                     | Analyze the field potentials due to static magnetic fields.                                                                          |
| C212.4                                                                                                                                                     | Explain how materials affect electric and magnetic fields.                                                                           |
| C212 .5                                                                                                                                                    | Analyze the relation between the fields under time varying situations and apply Maxwell's equations to electric and magnetic fields. |
| <b>Course Code &amp; Title : EC6404 Linear Integrated Circuits</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C213</b> |                                                                                                                                      |
| <b>Couse Index</b>                                                                                                                                         | <b>Course Outcomes</b>                                                                                                               |
| C213.1                                                                                                                                                     | Design linear and non-linear applications of op-amps.                                                                                |
| C213.2                                                                                                                                                     | Design applications using Analog multipliers and PLL.                                                                                |
| C213.3                                                                                                                                                     | Design ADC and DAC using op-amps.                                                                                                    |
| C213.4                                                                                                                                                     | Design waveform Generators using op-amps.                                                                                            |

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| C213.5                                                                                                                                                                               | Analyze special function ICs.                                                                                                      |
| <b>Course Code &amp; Title : EC6405 Control System Engineering</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C214</b>                           |                                                                                                                                    |
| <b>Couse Index</b>                                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                             |
| C214.1                                                                                                                                                                               | Perform modelling of control system using various techniques                                                                       |
| C214.2                                                                                                                                                                               | Obtain the time response and steady state error of control systems                                                                 |
| C214.3                                                                                                                                                                               | Design compensators and to plot the frequency response using various techniques                                                    |
| C214.4                                                                                                                                                                               | Determine the stability of control systems                                                                                         |
| C214.5                                                                                                                                                                               | Analyse and obtain state space model using state variables                                                                         |
| <b>Course Code &amp; Title : EC6411 Circuit and Simulation Integrated Laboratory</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C215</b>         |                                                                                                                                    |
| <b>Couse Index</b>                                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                             |
| C215.1                                                                                                                                                                               | Analyze various types of feedback amplifiers                                                                                       |
| C215.2                                                                                                                                                                               | Design oscillators, tuned amplifiers, wave-shaping circuits and multivibrators.                                                    |
| C215.3                                                                                                                                                                               | Design and simulate feedback amplifiers, oscillators, tuned amplifiers, wave-shaping circuits and multivibrators using SPICE tool. |
| <b>Course Code &amp; Title : EC6412 Linear Integrated Circuit Laboratory</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C216</b>                 |                                                                                                                                    |
| <b>Couse Index</b>                                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                             |
| C216.1                                                                                                                                                                               | Analyze the basics of linear integrated circuits and available ICs.                                                                |
| C216.2                                                                                                                                                                               | Design the oscillators, amplifiers and filters using operational amplifiers.                                                       |
| C216.3                                                                                                                                                                               | Analyze and implement the frequency multiplier using PLL.                                                                          |
| C216.4                                                                                                                                                                               | Design the oscillators and Multi vibrators using SPICE.                                                                            |
| C216.5                                                                                                                                                                               | Analyze the performance of DC power supply using ICs.                                                                              |
| <b>Course Code &amp; Title : EE6461 Electrical Engineering and Control System Laboratory</b><br><b>Semester &amp; Year of study : IV &amp; 2016-17</b><br><b>Course Index : C217</b> |                                                                                                                                    |

| <b>Couse Index</b>                                                                                                                                                     | <b>Course Outcomes</b>                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| C217.1                                                                                                                                                                 | Perform experiments to study the load characteristics of DC Motors/ Generators.         |
| C217.2                                                                                                                                                                 | Design bridge network circuit to measure the values of passive component.               |
| C217.3                                                                                                                                                                 | Analyse the stability of linear system through simulation software                      |
| C217.4                                                                                                                                                                 | Obtain transfer function of DC Generator.                                               |
| <b>III Year</b>                                                                                                                                                        |                                                                                         |
| <b>Course Code &amp; Title : EC6501 Digital Communication</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C301</b>                   |                                                                                         |
| <b>Couse Index</b>                                                                                                                                                     | <b>Course Outcomes</b>                                                                  |
| C301.1                                                                                                                                                                 | Design applications using the various types of sampling and quantization.               |
| C301.2                                                                                                                                                                 | Describe the various waveform coding schemes.                                           |
| C301.3                                                                                                                                                                 | Analyze the various baseband transmission schemes.                                      |
| C301.4                                                                                                                                                                 | Explain the various digital modulation schemes.                                         |
| C301.5                                                                                                                                                                 | Apply the various channel coding in digital communication systems.                      |
| <b>Course Code &amp; Title : EC6502 Principles of Digital Signal Processing</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C302</b> |                                                                                         |
| <b>Couse Index</b>                                                                                                                                                     | <b>Course Outcomes</b>                                                                  |
| C302.1                                                                                                                                                                 | Apply DFT for the analysis of digital signals & systems.                                |
| C302.2                                                                                                                                                                 | Design Infinite Impulse response (IIR) digital filters.                                 |
| C302.3                                                                                                                                                                 | Design Finite Impulse response (FIR) digital filters.                                   |
| C302.4                                                                                                                                                                 | Analyze the finite Word length effects in digital filters.                              |
| C302.5                                                                                                                                                                 | Explain the concept of Multirate Signal processing and Application of Adaptive Filters. |
| <b>Course Code &amp; Title : EC6503 Transmission Lines and Wave Guides</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C303</b>      |                                                                                         |
| <b>Couse</b>                                                                                                                                                           | <b>Course Outcomes</b>                                                                  |

| <b>Index</b>                                                                                                                                                         |                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| C303.1                                                                                                                                                               | Analyze the various types of transmission lines and the losses associated.                                                                        |
| C303.2                                                                                                                                                               | Analyze different parameters and constraints in high frequency transmission of information.                                                       |
| C303.3                                                                                                                                                               | Apply the concept of impedance transformation and matching using Smith chart.                                                                     |
| C303.4                                                                                                                                                               | Use the fundamental concept of filter theories and design it.                                                                                     |
| C303.5                                                                                                                                                               | Apply the fundamental concept of transmission in different guided systems and Utilize cavity resonators.                                          |
| <b>Course Code &amp; Title : GE6351 Environmental Science and Engineering</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C304</b> |                                                                                                                                                   |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                                            |
| C304.1                                                                                                                                                               | Summarize the importance of public awareness on environment and nature of biodiversity.                                                           |
| C304.2                                                                                                                                                               | Describe the various causes, effect and control measures of environmental pollution.                                                              |
| C304.3                                                                                                                                                               | Discuss the human development that leads to environmental disasters, the value of natural resources and their conservation.                       |
| C304.4                                                                                                                                                               | Explain the value of public participation in environmental protection, Environmental Management and Legislation Acts and sustainable development. |
| C304.5                                                                                                                                                               | Review the problems related to population and their remedial measures.                                                                            |
| <b>Course Code &amp; Title : EC6504 Microprocessor and Microcontroller</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C305</b>    |                                                                                                                                                   |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                                                            |
| C305.1                                                                                                                                                               | Write and implement programs on 8086 microprocessor.                                                                                              |
| C305.2                                                                                                                                                               | Implement the system bus structure of 8086 and coprocessor.                                                                                       |
| C305.3                                                                                                                                                               | Describe the I/O devices, peripherals and bus interfacing.                                                                                        |
| C305.4                                                                                                                                                               | Elaborate the operation of 8051 microcontroller architecture and implement ALP using 8051 instructions.                                           |
| C305.5                                                                                                                                                               | Design and implement 8051 microcontroller based systems.                                                                                          |
| <b>Course Code &amp; Title : EC6511 Digital Signal Processing Laboratory</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C306</b>  |                                                                                                                                                   |
| <b>Couse</b>                                                                                                                                                         | <b>Course Outcomes</b>                                                                                                                            |

| <b>Index</b>                                                                                                                                                                 |                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| C306.1                                                                                                                                                                       | Implement Linear and circular convolution programs using MATLAB and DSP Processor               |
| C306.2                                                                                                                                                                       | Implement FIR and IIR Filters Using MATLAB and DSP Processor.                                   |
| C306.3                                                                                                                                                                       | Implement Multi rate filters for various applications of DSP.                                   |
| C306.4                                                                                                                                                                       | Analyze the performance of DSP Processor.                                                       |
| C306.5                                                                                                                                                                       | Generate the Waveform of signals using DSP Processor.                                           |
| <b>Course Code &amp; Title : EC6512 Communication System Laboratory</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C307</b>               |                                                                                                 |
| <b>Couse Index</b>                                                                                                                                                           | <b>Course Outcomes</b>                                                                          |
| C307.1                                                                                                                                                                       | Analyze the effects of sampling and TDM.                                                        |
| C307.2                                                                                                                                                                       | Design AM & FM modulation and demodulation.                                                     |
| C307.3                                                                                                                                                                       | Implement Pulse Code Modulation and Delta Modulation.                                           |
| C307.4                                                                                                                                                                       | Implement the signal constellations of BPSK, QPSK and QAM.                                      |
| C307.5                                                                                                                                                                       | Demonstrate base band signaling schemes through implementation of FSK, PSK and DPSK.            |
| C307.6                                                                                                                                                                       | Implement Equalization algorithms and Error control coding schemes.                             |
| <b>Course Code &amp; Title : EC6513 Microprocessor and Microcontroller Laboratory</b><br><b>Semester &amp; Year of study : V &amp; 2017-18</b><br><b>Course Index : C308</b> |                                                                                                 |
| <b>Couse Index</b>                                                                                                                                                           | <b>Course Outcomes</b>                                                                          |
| C308.1                                                                                                                                                                       | Write ALP program for arithmetic operation, logical operations and data movement.               |
| C308.2                                                                                                                                                                       | Implement code conversion, decimal arithmetic and matrix operations.                            |
| C308.3                                                                                                                                                                       | Execute ALP program for floating point operations, string manipulations, sorting and searching. |
| C308.4                                                                                                                                                                       | Implement peripherals and interfacing with 8086 microprocessor.                                 |
| C308.5                                                                                                                                                                       | Write and execute ALP program for arithmetic, logical operations, square and cube program.      |
| <b>Course Code &amp; Title : MG6851 Principles of Management</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C309</b>                     |                                                                                                 |



| <b>Couse Index</b>                                                                                                                                    | <b>Course Outcomes</b>                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| C309.1                                                                                                                                                | Discuss the evolution of management, functions and roles of managers                                                   |
| C309.2                                                                                                                                                | Explain the different types of planning process and tools used for planning                                            |
| C309.3                                                                                                                                                | Elaborate different organization structures and functions of human resources manager                                   |
| C309.4                                                                                                                                                | Illustrate the different theories of motivation and leadership                                                         |
| C309.5                                                                                                                                                | Describe the control techniques and the role of technology in management                                               |
| <b>Course Code &amp; Title : CS6303 Computer Architecture</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C310</b> |                                                                                                                        |
| <b>Couse Index</b>                                                                                                                                    | <b>Course Outcomes</b>                                                                                                 |
| C310.1                                                                                                                                                | Analyze the performance of the computer operations and understand the instructions in MIPS architecture.               |
| C310.2                                                                                                                                                | Design arithmetic and logic unit.                                                                                      |
| C310.3                                                                                                                                                | Design a pipeline for execution of instructions with minimum hazards.                                                  |
| C310.4                                                                                                                                                | Determine the ways to take advantage of instruction level parallelism for high performance processor design.           |
| C310.5                                                                                                                                                | Explain the memory hierarchy, input/output mechanism and evaluate the performance of the memory system.                |
| <b>Course Code &amp; Title : CS6551 Computer Networks</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C311</b>     |                                                                                                                        |
| <b>Couse Index</b>                                                                                                                                    | <b>Course Outcomes</b>                                                                                                 |
| C311.1                                                                                                                                                | Summarize the concepts of the network fundamentals and different layers.                                               |
| C311.2                                                                                                                                                | Identify the components required to build different types of networks and to understand the internetworking protocols. |
| C311.3                                                                                                                                                | Explore the network with routing and multicasting.                                                                     |
| C311.4                                                                                                                                                | Trace the flow of information from one node to another in the networks.                                                |
| C311.5                                                                                                                                                | Analyze the features and operations of various application layer protocols such as HTTP, DNS, and SMTP.                |

| <b>Course Code &amp; Title : EC6601 VLSI Design</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C312</b>                      |                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Couse Index</b>                                                                                                                                               | <b>Course Outcomes</b>                                                                                                     |
| C312.1                                                                                                                                                           | Explain the basic characteristics of MOSFET and different CMOS process technologies.                                       |
| C312.2                                                                                                                                                           | Design and analyze the various combinational logic circuits.                                                               |
| C312.3                                                                                                                                                           | Model the sequential digital system using Hardware Description Language.                                                   |
| C312.4                                                                                                                                                           | Design arithmetic building blocks.                                                                                         |
| C312 .5                                                                                                                                                          | Apply the techniques of chip design using FPGA programmable devices.                                                       |
| <b>Course Code &amp; Title : EC6602 Antenna and Wave propagation</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C313</b>     |                                                                                                                            |
| <b>Couse Index</b>                                                                                                                                               | <b>Course Outcomes</b>                                                                                                     |
| C313.1                                                                                                                                                           | Analyze the basic antenna parameters and radiations from dipole antennas.                                                  |
| C313.2                                                                                                                                                           | Describe the various types of aperture and slot antennas.                                                                  |
| C313.3                                                                                                                                                           | Explain about the various kinds of antenna arrays.                                                                         |
| C313.4                                                                                                                                                           | Apply the various types of special and modern antennas based on the applications.                                          |
| C313.5                                                                                                                                                           | Analyze the different types of propagation of radio waves at different frequencies.                                        |
| <b>Course Code &amp; Title : EC6001 Medical Electronics - Elective I</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C314</b> |                                                                                                                            |
| <b>Couse Index</b>                                                                                                                                               | <b>Course Outcomes</b>                                                                                                     |
| C314.1                                                                                                                                                           | Explain about the physiological parameters and recording methods.                                                          |
| C314.2                                                                                                                                                           | Analyze the bio-chemicals and various physiological information.                                                           |
| C314.3                                                                                                                                                           | Describe various assist devices used in hospitals.                                                                         |
| C314.4                                                                                                                                                           | Explain the equipment used for physical medicine and the various recently developed diagnostic and therapeutic techniques. |
| C314.5                                                                                                                                                           | Apply the concepts of medical Instrumentation in recent technology (laser, telemedicine, endoscopy unit).                  |

| <b>Course Code &amp; Title : EC6611 Computer Networks Laboratory</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C315</b>                     |                                                                                                                              |
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| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                                                       |
| C315.1                                                                                                                                                                           | Perform client-server communication between two desktop computers using Socket Programming.                                  |
| C315.2                                                                                                                                                                           | Implement the different protocols.                                                                                           |
| C315.3                                                                                                                                                                           | Simulate various network topologies like Star, Bus and Ring.                                                                 |
| C315.4                                                                                                                                                                           | Implement and compare the different routing algorithms.                                                                      |
| C315.5                                                                                                                                                                           | Simulate the algorithms with the help of Network Simulator tool.                                                             |
| <b>Course Code &amp; Title : EC6612 VLSI Design Laboratory</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C316</b>                           |                                                                                                                              |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                                                       |
| C316.1                                                                                                                                                                           | Write HDL code for basic as well as advanced digital integrated circuits.                                                    |
| C316.2                                                                                                                                                                           | Synthesize, Place and Route the digital circuits.                                                                            |
| C316.3                                                                                                                                                                           | Import the logic modules in to FPGA boards.                                                                                  |
| C316.4                                                                                                                                                                           | Design and Simulate the digital circuits using EDA platforms.                                                                |
| C316.5                                                                                                                                                                           | Design and Simulate the CMOS transistor differential amplifiers using EDA tools.                                             |
| <b>Course Code &amp; Title : GE6674 Communication and Soft Skills - Laboratory Based</b><br><b>Semester &amp; Year of study : VI &amp; 2017-18</b><br><b>Course Index : C317</b> |                                                                                                                              |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                                                       |
| C317.1                                                                                                                                                                           | Identify and interpret visuals, communicate in formal and informal conversations, give presentations, and participate in GD. |
| C317.2                                                                                                                                                                           | Explain reading comprehension passages of higher levels, draft Resume, cover letter, reports, emails, and write blogs.       |
| C317.3                                                                                                                                                                           | Differentiate between IELTS & TOEFL and take placement oriented verbal ability tests.                                        |
| C317.4                                                                                                                                                                           | Demonstrate appropriate verbal, non-verbal and paralinguistic skills while attending job interviews.                         |
| C317.5                                                                                                                                                                           | Show leadership traits, team skills and essential soft skills and efficiency to excel as a professional.                     |

| IV Year                                                                                                                                                             |                                                                                                              |
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| <b>Course Code &amp; Title : EC6701 RF and Microwave Engineering</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C401</b>       |                                                                                                              |
| Couse Index                                                                                                                                                         | Course Outcomes                                                                                              |
| C401.1                                                                                                                                                              | Design and analyse multiport RF networks                                                                     |
| C401.2                                                                                                                                                              | Formulate solutions for stability and impedance matching issues in microwave amplifiers                      |
| C401.3                                                                                                                                                              | Explain the basic concept of various microwave devices                                                       |
| C401.4                                                                                                                                                              | Elaborate the concepts involved in Microwave generation                                                      |
| C401.5                                                                                                                                                              | Measure and analyse Microwave signal and its parameters                                                      |
| <b>Course Code &amp; Title : EC6702 Optical Communication and Networks</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C402</b> |                                                                                                              |
| Couse Index                                                                                                                                                         | Course Outcomes                                                                                              |
| C402.1                                                                                                                                                              | Fundamental concept of optical fiber modes and their configurations.                                         |
| C402.2                                                                                                                                                              | Analyse the various signal degradation factors associated with optical fiber.                                |
| C402.3                                                                                                                                                              | Explain the various optical sources and optical detectors and their use in the optical communication system. |
| C402.4                                                                                                                                                              | Describe the techniques required to measure the optical fiber systems.                                       |
| C402.5                                                                                                                                                              | Analyse the Digital Transmission and its associated parameters on system performance                         |
| <b>Course Code &amp; Title : EC6703 Embedded and Real Time Systems</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C403</b>     |                                                                                                              |
| Couse Index                                                                                                                                                         | Course Outcomes                                                                                              |
| C403.1                                                                                                                                                              | Explain the fundamental concepts of designing Embedded and Real Time Systems                                 |
| C403.2                                                                                                                                                              | Describe the computing required for Real Time Embedded Systems.                                              |
| C403.3                                                                                                                                                              | Summarize the concepts of scheduling in Real Time Operating System.                                          |
| C403.4                                                                                                                                                              | Analyse the techniques required to create complex Embedded System.                                           |

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| C403.5                                                                                                                                                    | Apply the concepts of Embedded System Design in creating the model Real Time applications.                                                                 |
| <b>Course Code &amp; Title : EC6004 Satellite Communication</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C404</b>  |                                                                                                                                                            |
| <b>Couse Index</b>                                                                                                                                        | <b>Course Outcomes</b>                                                                                                                                     |
| C404.1                                                                                                                                                    | Determine the azimuth and elevation angles and visibility of a geostationary satellite from an earth station.                                              |
| C404.2                                                                                                                                                    | Explain the concept of signal propagation of space segment components and create link budgets for an uplink and a downlink,                                |
| C404.3                                                                                                                                                    | Analyze the effect of rain attenuation in a satellite link and the availability of the link based on geographic location of the earth terminals.           |
| C404.4                                                                                                                                                    | Design satellite communication system to carry voice, video or data signals using analog or digital modulation.                                            |
| C404.5                                                                                                                                                    | Analyze the various types of satellite services according to its applications.                                                                             |
| <b>Course Code &amp; Title : IT6005 Digital Image Processing</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C405</b> |                                                                                                                                                            |
| <b>Couse Index</b>                                                                                                                                        | <b>Course Outcomes</b>                                                                                                                                     |
| C405.1                                                                                                                                                    | Analyse the fundamental concepts of a digital image processing system                                                                                      |
| C405.2                                                                                                                                                    | Examine various types of images, intensity transformations and spatial filtering and to develop Fourier transform for image processing in frequency domain |
| C405.3                                                                                                                                                    | Determine the methodologies for image segmentation, restoration etc                                                                                        |
| C405.4                                                                                                                                                    | Apply various coding methods used for Image Compression                                                                                                    |
| C405.5                                                                                                                                                    | Explain about Image Restoration and Recognition techniques                                                                                                 |
| <b>Course Code &amp; Title : EC6007 Speech Processing</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C406</b>        |                                                                                                                                                            |
| <b>Couse Index</b>                                                                                                                                        | <b>Course Outcomes</b>                                                                                                                                     |
| C406.1                                                                                                                                                    | Explain the fundamentals and classification of speech sounds.                                                                                              |
| C406.2                                                                                                                                                    | Analyze, extract and compare the various speech parameters.                                                                                                |
| C406.3                                                                                                                                                    | Apply an appropriate speech model for a given application.                                                                                                 |
| C406.4                                                                                                                                                    | Explain the various speech recognition systems.                                                                                                            |

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| C406.5                                                                                                                                                                           | Apply different speech synthesis techniques depending upon the classification of speech parameters. |
| <b>Course Code &amp; Title : EC6011 Electro Magnetic Interference and Compatibility</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C407</b> |                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                              |
| C407.1                                                                                                                                                                           | Find solution to EMI Sources, EMI problems in PCB level / Subsystem and system level design.        |
| C407.2                                                                                                                                                                           | Measure emission immunity level from different systems to couple with the prescribed EMC standards. |
| C407.3                                                                                                                                                                           | Analyze the Mitigation techniques.                                                                  |
| C407.4                                                                                                                                                                           | Apply the data of EMI tests, and check the coupling and Grounding tests.                            |
| C407.5                                                                                                                                                                           | Apply the various EMI test methods.                                                                 |
| <b>Course Code &amp; Title : EC6013 Advanced Microprocessors and Microcontrollers</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C408</b>   |                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                              |
| C408.1                                                                                                                                                                           | Describe the operation of Pentium processor and programming concepts.                               |
| C408.2                                                                                                                                                                           | Analyze the operation of ARM processor, instruction sets and development tools.                     |
| C408.3                                                                                                                                                                           | Develop DSP applications using ARM processor.                                                       |
| C408.4                                                                                                                                                                           | Explain about MOTOROLA 68HC11 architecture and instruction sets.                                    |
| C408.5                                                                                                                                                                           | Elaborate the operation of PIC microcontroller and instruction sets                                 |
| <b>Course Code &amp; Title : EC6016 Opto Electronic devices</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C409</b>                         |                                                                                                     |
| <b>Couse Index</b>                                                                                                                                                               | <b>Course Outcomes</b>                                                                              |
| C409.1                                                                                                                                                                           | Explain the wave nature of light and basics of solid state physics.                                 |
| C409.2                                                                                                                                                                           | Elucidate the operation of display devices and laser.                                               |
| C409.3                                                                                                                                                                           | Describe the various optical detection devices.                                                     |
| C409.4                                                                                                                                                                           | Apply concept of optoelectronic modulator principle in fiber optic communication.                   |

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| C409.5                                                                                                                                                            | Describe the integration of Optoelectronic devices on a chip.                                                                    |
| <b>Course Code &amp; Title : EC6711 Embedded Laboratory</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C410</b>              |                                                                                                                                  |
| <b>Couse Index</b>                                                                                                                                                | <b>Course Outcomes</b>                                                                                                           |
| C410.1                                                                                                                                                            | Write programs in ARM for a specific Application                                                                                 |
| C410.2                                                                                                                                                            | Interface memory with ARM processor and write a program related to memory operations                                             |
| C410.3                                                                                                                                                            | Interface A/D and D/A convertors with ARM system                                                                                 |
| C410.4                                                                                                                                                            | Analyze the performance of interrupt                                                                                             |
| C410.5                                                                                                                                                            | Write programs for interfacing keyboard, display, motor and sensor                                                               |
| <b>Course Code &amp; Title : EC6712 Optical and Microwave Laboratory</b><br><b>Semester &amp; Year of study : VII &amp; 2018-19</b><br><b>Course Index : C411</b> |                                                                                                                                  |
| <b>Couse Index</b>                                                                                                                                                | <b>Course Outcomes</b>                                                                                                           |
| C411.1                                                                                                                                                            | Analyze the performance of simple optical link.                                                                                  |
| C411.2                                                                                                                                                            | Test microwave and optical components.                                                                                           |
| C411.3                                                                                                                                                            | Determine the losses in fiber.                                                                                                   |
| C411.4                                                                                                                                                            | Analyze the mode characteristics of fiber.                                                                                       |
| C411.5                                                                                                                                                            | Analyze the radiation pattern of antenna.                                                                                        |
| <b>Course Code &amp; Title : EC6801 Wireless Communication</b><br><b>Semester &amp; Year of study : VIII &amp; 2018-19</b><br><b>Course Index : C412</b>          |                                                                                                                                  |
| <b>Couse Index</b>                                                                                                                                                | <b>Course Outcomes</b>                                                                                                           |
| C412.1                                                                                                                                                            | Explain the modelling of various wireless channels.                                                                              |
| C412.2                                                                                                                                                            | Apply the various cellular concepts like frequency reuse, channel assignments, handoff strategies etc., in mobile communication. |
| C412.3                                                                                                                                                            | Analyze the performance of various digital signalling schemes of fading channels.                                                |
| C412.4                                                                                                                                                            | Apply the multipath mitigation techniques based on the application.                                                              |

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| C412 .5                                                                                                                                                              | Implement the concept of transmit/receive diversity in MIMO systems.                                         |
| <b>Course Code &amp; Title : EC6802 Wireless Networks</b><br><b>Semester &amp; Year of study : VIII &amp; 2018-19</b><br><b>Course Index : C413</b>                  |                                                                                                              |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                       |
| C413.1                                                                                                                                                               | Analyze different standards used for Wireless LAN.                                                           |
| C413.2                                                                                                                                                               | Explain the Network Layer protocols used for Mobile Applications.                                            |
| C413.3                                                                                                                                                               | Describe various techniques in Mobile Transport Layers.                                                      |
| C413.4                                                                                                                                                               | Implement different type of applications for smart phones and mobile devices with latest network strategies. |
| C413.5                                                                                                                                                               | Converse with the latest 3G/4G and WiMAX Networks and its Architecture.                                      |
| <b>Course Code &amp; Title : GE6075 Professional Ethics in Engineering</b><br><b>Semester &amp; Year of study : VIII &amp; 2018-19</b><br><b>Course Index : C414</b> |                                                                                                              |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                       |
| C414.1                                                                                                                                                               | Apply ethics, morals and human values in society. The students will be able to                               |
| C414.2                                                                                                                                                               | Explain about engineering ethics.                                                                            |
| C414.3                                                                                                                                                               | Describe the responsibilities of engineers as experimenters.                                                 |
| C414.4                                                                                                                                                               | Analyze the safety, risks, risk benefit analysis and rights of an engineer.                                  |
| C414.5                                                                                                                                                               | Discuss the Importance of the global issues, moral leadership and code of conduct.                           |
| <b>Course Code &amp; Title : GE6757 Total Quality Management</b><br><b>Semester &amp; Year of study : VIII &amp; 2018-19</b><br><b>Course Index : C415</b>           |                                                                                                              |
| <b>Couse Index</b>                                                                                                                                                   | <b>Course Outcomes</b>                                                                                       |
| C415.1                                                                                                                                                               | Discuss the contributions of Quality Guru.                                                                   |
| C415.2                                                                                                                                                               | Explain the principles of TQM.                                                                               |
| C415.3                                                                                                                                                               | Apply the tools and techniques of quality management to manufacturing and service processes.                 |
| C415.4                                                                                                                                                               | Describe TQM tools and techniques such as Control Charts, QFD and TPM.                                       |



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| C415.5                                                                                                                                                              | Discuss the elements of Quality system standards.                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course Code &amp; Title : CS6701 Cryptography and Network Security</b><br><b>Semester &amp; Year of study : VIII &amp; 2018-19</b><br><b>Course Index : C416</b> |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Couse Index</b>                                                                                                                                                  | <b>Course Outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                       |
| C416.1                                                                                                                                                              | Explain the basic concepts of OSI security architecture, Finite Fields and Number Theory.                                                                                                                                                                                                                                                                                                                                    |
| C416.2                                                                                                                                                              | Compare the various Cryptographic Techniques.                                                                                                                                                                                                                                                                                                                                                                                |
| C416.3                                                                                                                                                              | Determine the usage of Hash functions and Digital signature.                                                                                                                                                                                                                                                                                                                                                                 |
| C416.4                                                                                                                                                              | Design the various secure applications.                                                                                                                                                                                                                                                                                                                                                                                      |
| C416.5                                                                                                                                                              | Inject Security coding in the developed applications.                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Code &amp; Title : EC6811 Project Work</b><br><b>Semester &amp; Year of study : VIII &amp; 2018-19</b><br><b>Course Index : C417</b>                      |                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Couse Index</b>                                                                                                                                                  | <b>Course Outcomes</b>                                                                                                                                                                                                                                                                                                                                                                                                       |
| C417.1                                                                                                                                                              | Apply the fundamental knowledge and skills, which are acquired within the technical area, to a given problem                                                                                                                                                                                                                                                                                                                 |
| C417.2                                                                                                                                                              | <ul style="list-style-type: none"> <li>Identify and summarize an appropriate list of literature review, analyze previous researchers' work and relate them to the project</li> <li>Within given constraints, even with limited information, the students will be able to independently analyze and discuss complex inquiries/problems and handle larger problems on the advanced level within the technical area.</li> </ul> |
| C417.3                                                                                                                                                              | <ul style="list-style-type: none"> <li>Design engineering solutions to complex problems in a systematic approach.</li> <li>Identify and apply appropriate parameters, assumptions and design criteria in consideration of health and safety (example: the use of codes of practice), ethics, economics, environment, sustainability</li> </ul>                                                                               |
| C417.4                                                                                                                                                              | Apply research and conduct experiments, as well as to analyze and interpret data that yield the results and answer important applicable research questions.                                                                                                                                                                                                                                                                  |
| C417.5                                                                                                                                                              | Utilize technology tools for communication, collaboration, information management, and decision support.                                                                                                                                                                                                                                                                                                                     |
| C417.6                                                                                                                                                              | Demonstrate the knowledge, skills and attitudes of a professional engineer.                                                                                                                                                                                                                                                                                                                                                  |
| C417.7                                                                                                                                                              | Interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment.                                                                                                                                                                                                                                                                                                  |
| C417.8                                                                                                                                                              | Demonstrate a strong working knowledge of ethics and professional responsibility.                                                                                                                                                                                                                                                                                                                                            |
| C417.9                                                                                                                                                              | Document and present one's own work, for a given target group, with strict requirements on structure, format, and language usage.                                                                                                                                                                                                                                                                                            |

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| C417.10 | Present the project outlining the approach and expected results using good oral and written presentation skills. |
| C417.11 | Demonstrate effective organizational leadership and change skills for managing projects and project teams.       |
| C417.12 | Recognize the need for life-long learning by undergoing the project work.                                        |